

Incident Response Report

Title: Simulated SOC Log Analysis

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1. Executive Summary

This report documents the analysis of simulated Security Operations Center (SOC) logs. The investigation identified multiple indicators of compromise, including malware infections, credential abuse, lateral movement, and ransomware-like behavior.

Several internal hosts appear compromised, and multiple user accounts exhibit suspicious authentication patterns. Immediate containment and remediation actions are required.

2. Scope and Data Sources

2.1 Log Types Analyzed

- Network connection logs
- Authentication logs (successful and failed)
- File access logs
- Endpoint malware detection alerts

2.2 Environment Overview

- Internal IP ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
- External-facing IP activity observed
- Multiple shared user accounts

3. High-Level Timeline

- Initial network connection attempts
- Repeated failed authentication attempts
- Successful logins following failures
- Malware detections on multiple hosts
- File access and ransomware behavior indicators

4. Key Findings

4.1 High-Risk IP Addresses

IP Address	Observed Activity
203.0.113.77	Login failures, malware alerts, file access
172.16.0.3	Trojan, spyware, ransomware behavior
10.0.0.5	Rootkit and trojan detections
198.51.100.42	Malware combined with authentication events

4.2 Suspicious User Accounts

User	Indicators
bob	Malware alerts, ransomware behavior, credential misuse
david	Repeated connection attempts, malware detections
alice	Rootkit and spyware indicators
eve	File access following malware alerts

4.3 Malware Identified

- Trojan Detected
- Rootkit Signature
- Spyware Alert
- Worm Infection Attempt
- Ransomware Behavior

These detections indicate both initial compromise and post-exploitation activity.

5. Attack Pattern Assessment

The observed activity aligns with a multi-stage attack lifecycle:

- Initial Access** – Network connection attempts
- Credential Access** – Failed logins followed by success
- Execution & Persistence** – Malware installation
- Lateral Movement** – Activity across internal IP ranges
- Impact** – Ransomware-like behavior

6. Indicators of Compromise (IOCs)

6.1 IP Addresses

```
203.0.113.77
172.16.0.3
10.0.0.5
198.51.100.42
```

6.2 User Accounts

```
bob
david
alice
eve
```

7. Evidence and Supporting Screenshots

This section contains screenshots and supporting material used to validate the findings presented in this report. Each item corresponds to observations discussed in earlier sections.

Evidence 1: Event Timeline and Timestamps

Related Sections: 3 (High-Level Timeline)

Screenshots showing chronological system log entries, including timestamps that demonstrate the sequence of events from initial access through post-compromise activity.

New Search

source="SOC_Task2_Sample_Logs.txt" host="windows" sourcetype="misc"

Time range: during Thu, Jul 3, 2025

50 events (7/3/25 12:00:00.000 AM to 7/4/25 12:00:00.000 AM) No Event Sampling

Job

Smart Mode

Events (50)

Patterns

Statistics

Visualization

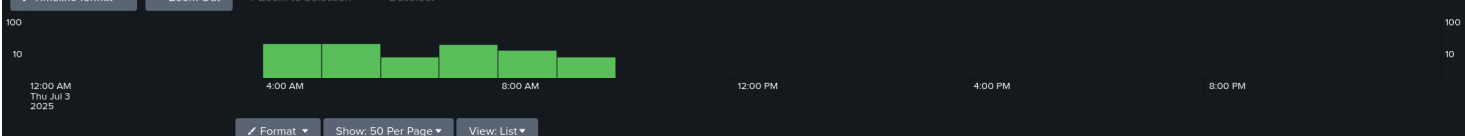
Timeline format

Zoom Out

Zoom to Selection

Deselect

1 hour per column



Format

Show: 50 Per Page

View: List

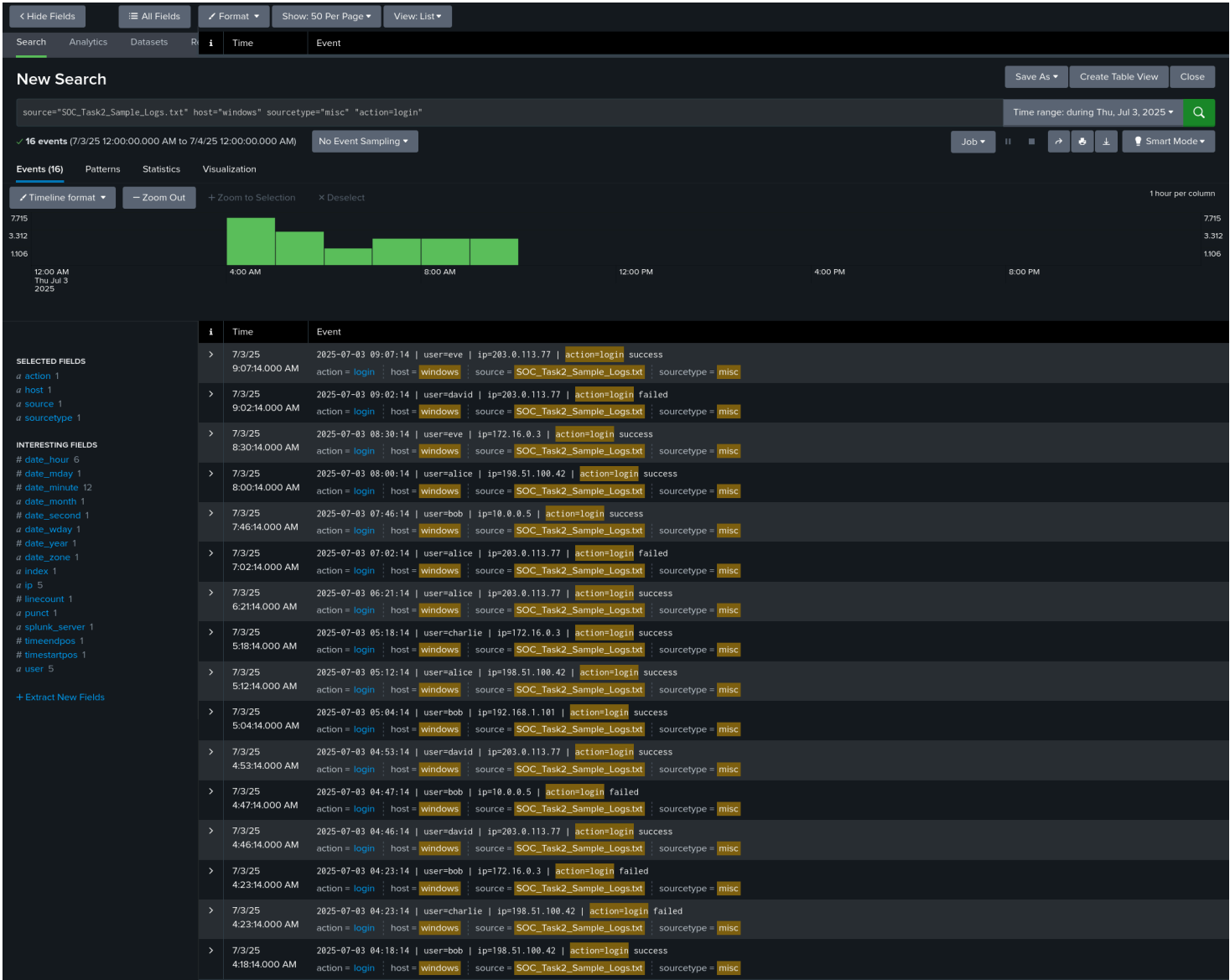
< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS a action 4 a host 1 a source 1 a sourcetype 1	INTERESTING FIELDS # date_hour 6 # date_mday 1 # date_minute 33 # date_month 1 # date_second 1 # date_wday 1 # date_year 1 a date_zone 1 a index 1 a ip 5 # linecount 1 a punct 3 a splunk_server 1 a threat 5 # timeendpos 1 # timestartpos 1 a user 5 + Extract New Fields		7/3/25 9:10:14.000 AM	2025-07-03 09:10:14 user=bob ip=172.16.0.3 action=malware detected threat=Ransomware Behavior action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 9:10:14.000 AM	2025-07-03 09:10:14 user=bob ip=198.51.100.42 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 9:07:14.000 AM	2025-07-03 09:07:14 user=eve ip=203.0.113.77 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 9:02:14.000 AM	2025-07-03 09:02:14 user=david ip=203.0.113.77 action=login failed action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:42:14.000 AM	2025-07-03 08:42:14 user=eve ip=172.16.0.3 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:42:14.000 AM	2025-07-03 08:42:14 user=charlie ip=203.0.113.77 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:31:14.000 AM	2025-07-03 08:31:14 user=eve ip=203.0.113.77 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:30:14.000 AM	2025-07-03 08:30:14 user=eve ip=172.16.0.3 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:21:14.000 AM	2025-07-03 08:21:14 user=david ip=172.16.0.3 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:20:14.000 AM	2025-07-03 08:20:14 user=charlie ip=192.168.1.101 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 8:00:14.000 AM	2025-07-03 08:00:14 user=alice ip=198.51.100.42 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:57:14.000 AM	2025-07-03 07:57:14 user=david ip=10.0.0.5 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:51:14.000 AM	2025-07-03 07:51:14 user=eve ip=10.0.0.5 action=malware detected threat=Rootkit Signature action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:46:14.000 AM	2025-07-03 07:46:14 user=bob ip=10.0.0.5 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:45:14.000 AM	2025-07-03 07:45:14 user=charlie ip=172.16.0.3 action=malware detected threat=Trojan Detected action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:44:14.000 AM	2025-07-03 07:44:14 user=bob ip=192.168.1.101 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:44:14.000 AM	2025-07-03 07:44:14 user=bob ip=203.0.113.77 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:38:14.000 AM	2025-07-03 07:38:14 user=charlie ip=172.16.0.3 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:36:14.000 AM	2025-07-03 07:36:14 user=david ip=10.0.0.5 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:22:14.000 AM	2025-07-03 07:22:14 user=charlie ip=192.168.1.101 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:18:14.000 AM	2025-07-03 07:18:14 user=bob ip=203.0.113.77 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 7:02:14.000 AM	2025-07-03 07:02:14 user=alice ip=203.0.113.77 action=login failed action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 6:21:14.000 AM	2025-07-03 06:21:14 user=alice ip=203.0.113.77 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 6:13:14.000 AM	2025-07-03 06:13:14 user=charlie ip=10.0.0.5 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 6:10:14.000 AM	2025-07-03 06:10:14 user=david ip=203.0.113.77 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 6:01:14.000 AM	2025-07-03 06:01:14 user=bob ip=172.16.0.3 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:49:14.000 AM	2025-07-03 05:49:14 user=charlie ip=192.168.1.101 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:48:14.000 AM	2025-07-03 05:48:14 user=bob ip=10.0.0.5 action=malware detected threat=Trojan Detected action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:45:14.000 AM	2025-07-03 05:45:14 user=david ip=172.16.0.3 action=malware detected threat=Trojan Detected action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:44:14.000 AM	2025-07-03 05:44:14 user=bob ip=198.51.100.42 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:42:14.000 AM	2025-07-03 05:42:14 user=eve ip=203.0.113.77 action=malware detected threat=Trojan Detected action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:33:14.000 AM	2025-07-03 05:33:14 user=david ip=198.51.100.42 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:30:14.000 AM	2025-07-03 05:30:14 user=eve ip=192.168.1.101 action=malware detected threat=Trojan Detected action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:27:14.000 AM	2025-07-03 05:27:14 user=david ip=203.0.113.77 action=connection attempt action = connection host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:18:14.000 AM	2025-07-03 05:18:14 user=charlie ip=172.16.0.3 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:12:14.000 AM	2025-07-03 05:12:14 user=alice ip=198.51.100.42 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:06:14.000 AM	2025-07-03 05:06:14 user=bob ip=203.0.113.77 action=malware detected threat=Worm Infection Attempt action = malware detected host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 5:04:14.000 AM	2025-07-03 05:04:14 user=bob ip=192.168.1.101 action=login success action = login host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 4:53:14.000 AM	2025-07-03 04:53:14 user=alice ip=203.0.113.77 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc
			7/3/25 4:53:14.000 AM	2025-07-03 04:53:14 user=alice ip=203.0.113.77 action=file accessed action = file host = windows source = SOC_Task2_Sample_Logs.txt sourcetype = misc

>	7/3/25 4:53:14.000 AM	2025-07-03 04:53:14 user=david ip=203.0.113.77 action=login success	action = login	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:47:14.000 AM	2025-07-03 04:47:14 user=bob ip=10.0.0.5 action=login failed	action = login	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:46:14.000 AM	2025-07-03 04:46:14 user=david ip=203.0.113.77 action=login success	action = login	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:41:14.000 AM	2025-07-03 04:41:14 user=alice ip=172.16.0.3 action=malware detected threat=Spyware Alert	action = malware detected	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:29:14.000 AM	2025-07-03 04:29:14 user=alice ip=192.168.1.101 action=malware detected threat=Trojan Detected	action = malware detected	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:27:14.000 AM	2025-07-03 04:27:14 user=david ip=172.16.0.3 action=connection attempt	action = connection	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:23:14.000 AM	2025-07-03 04:23:14 user=bob ip=172.16.0.3 action=login failed	action = login	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:23:14.000 AM	2025-07-03 04:23:14 user=charlie ip=198.51.100.42 action=login failed	action = login	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:19:14.000 AM	2025-07-03 04:19:14 user=david ip=10.0.0.5 action=connection attempt	action = connection	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:19:14.000 AM	2025-07-03 04:19:14 user=alice ip=198.51.100.42 action=malware detected threat=Rootkit Signature	action = malware detected	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc
>	7/3/25 4:18:14.000 AM	2025-07-03 04:18:14 user=bob ip=198.51.100.42 action=login success	action = login	host = windows	source = SOC_Task2_Sample_Logs.txt	sourcetype = misc

Evidence 2: Network Connection Logs

Related Sections: 4.1 (High-Risk IP Addresses)

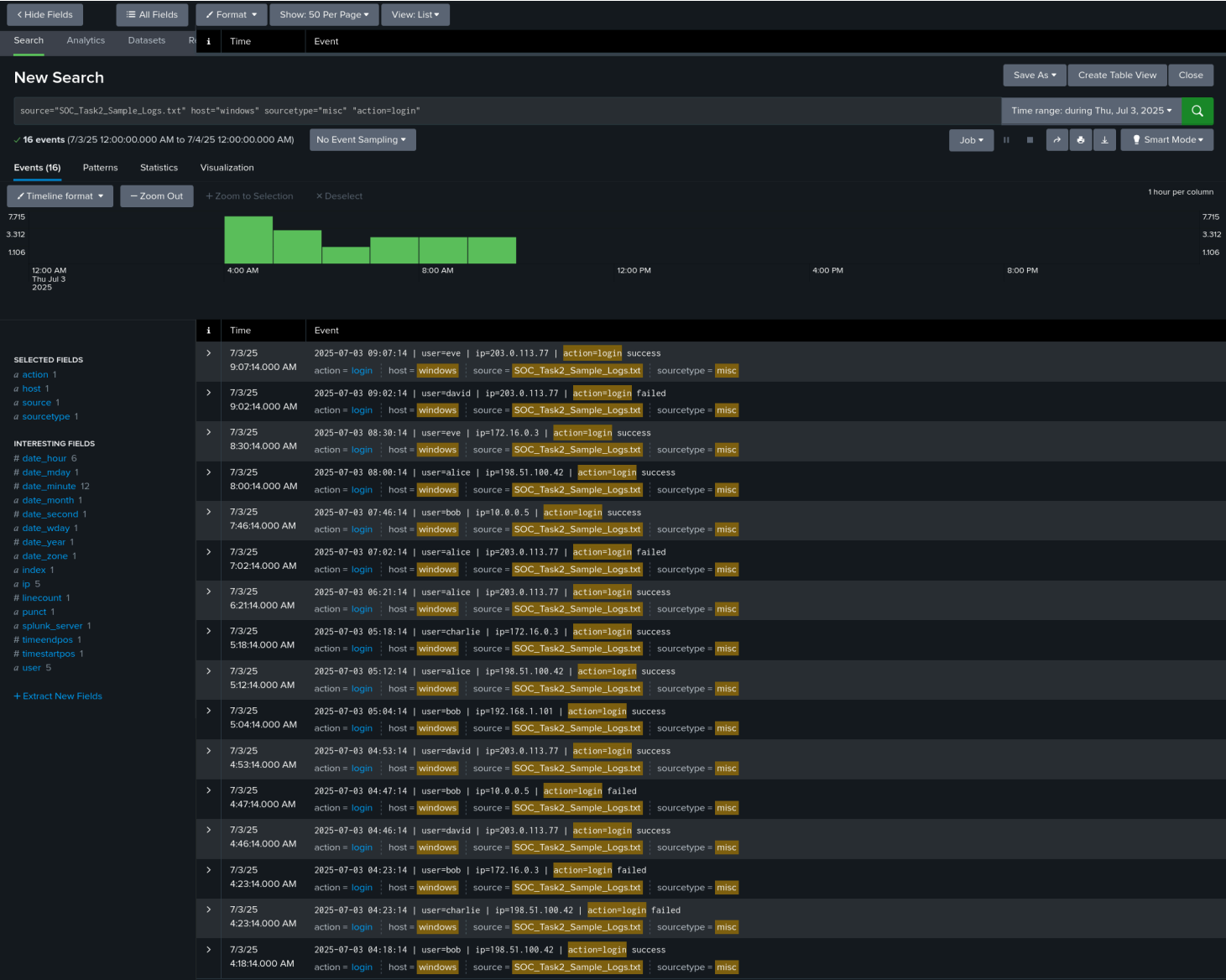
Screenshots highlighting repeated network connection attempts and suspicious activity associated with identified internal and external IP addresses.



Evidence 3: Authentication Logs

Related Sections: 4.2 (Suspicious User Accounts)

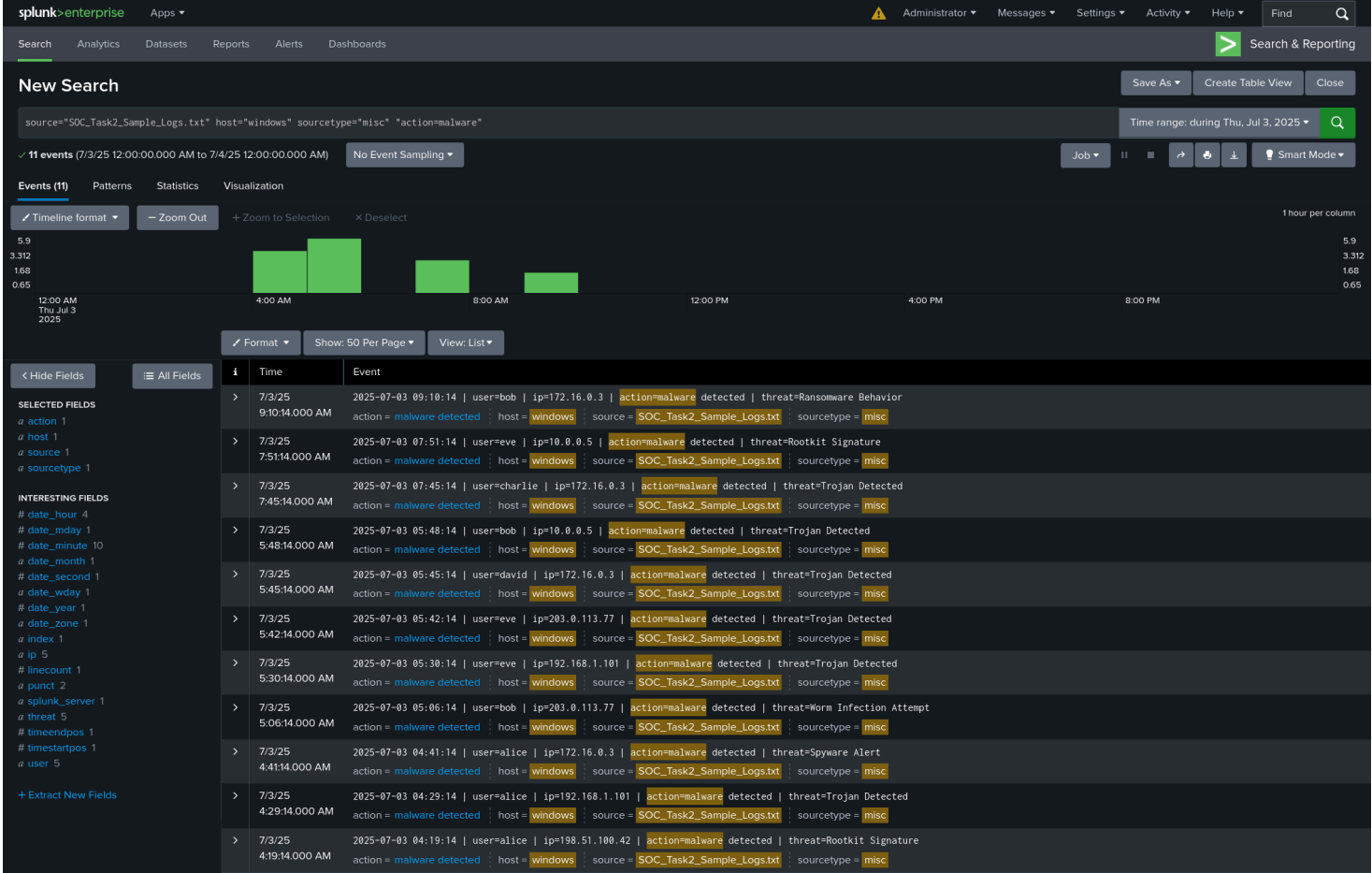
Screenshots showing failed and successful login attempts, including cases where successful authentication followed multiple failures, indicating potential credential compromise.



Evidence 4: Malware Detection Alerts

Related Sections: 4.3 (Malware Identified)

Screenshots of endpoint detection alerts showing malware classifications, affected hosts, and associated user accounts.



8. Recommendations

8.1 Immediate Actions

- Isolate compromised hosts
- Disable and reset affected user credentials
- Block malicious IP addresses at the firewall

8.2 Short-Term Improvements

- Perform full EDR scans
- Review authentication logs for credential abuse
- Improve SIEM correlation rules

8.3 Long-Term Mitigations

- Enforce multi-factor authentication (MFA)
- Improve internal network segmentation
- Implement behavioral-based detection controls

9. Conclusion

The analysis confirms an active security incident involving malware infections, credential compromise, and potential ransomware deployment.

Prompt containment and remediation are required to prevent further impact.

End of Report